Explain the project to your professor

How it was made ů Maths ů Graphs ů Terms ů Books

App: `https://che331.streamlit.app` ů Report: `docs/report/report.pdf` ů Tests: GitHub / tests

Pitch

CHE331's four reactor labs are run under fixed conditions and analysed later. ReactorGenome is a laptop web app beside the rig: type the same readings, get live fits and diagnostics, switch assumptions, then use your fitted k in a Design Studio the lab cannot build. No new sensors.

1 How the app was made (basics — no code deep dive)

1.1 What you actually built

A **web application**: open a browser, no separate desktop install. One site serves Batch, CSTR, PFR, RTD and a Design Studio. Live URL: `https://che331.streamlit.app`.

1.2 Tools (if they ask "what did you use?")

Piece	Role in plain words
Python	Language the whole project is written in.
Streamlit	Turns Python into a clickable web page (inputs, tables, tabs).
NumPy / SciPy	Numbers, fitting curves, solving ODEs (reactor profiles, Arrhenius, RTD).
Pandas	Readings tables (rows of T, flow, conductivity).
Plotly	Interactive graphs (zoom, hover, save PNG).
HTML/JS board	Drag-and-drop plant drawing in Design Studio.
Streamlit Cloud	Free hosting so anyone with the link can open the app.

1.3 Architecture (one picture in words)

- **Engine:** the chemistry and reactor equations only (no buttons). CSTR, PFR, batch, RTD, flowsheet solver.
- **Pages:** what the student sees — one page per experiment + Design Studio.
- Same layout every experiment: **left** assumptions, **middle** live reactor, **right** results, **below** readings table, **tabs** analysis and export.

1.4 How a number reaches the screen

Student types conductivity → app converts to conversion X → builds the design-plot coordinate y → fits a line to get k at that T → Arrhenius across temperatures → E_a and predictions. Every switch on the left changes the model and redraws.

1.5 How you know it is correct

Analytic limits (ideal CSTR/PFR, RTD moments, recycle) plus replaying 2026 lab-report sheets with their own conventions (k and E_a match to four figures). Test suite: GitHub `tests` folder.

2 Reference books and course material

Say these names if asked "where do the models come from?"

Source	What you took from it
H. S. Fogler, *Elements of Chemical Reaction Engineering*	Mole balances, CSTR/PFR design equations, Arrhenius, non-ideal reactors, energy balance / multiple steady states, Levenspiel plots, RTD (E, F, t_m, σ^2, tanks-in-series, dispersion / Pe).
O. Levenspiel, *Chemical Reaction Engineering*	Same reactor types; classic X–Da comparison; Levenspiel plot (area = size).
CHE331 lab manuals (Batch, CSTR, PFR, RTD)	Observation-table layout, design-plot conventions, typed Λ_∞, how groups report k and E_a.
Ionic conductivity data (limiting molar conductivities)	Default $\Lambda_\infty \approx 0.366\,\Lambda_0$ (OH^- replaced by acetate).

Say this

"I did not invent new reactor theory. I implemented the Fogler / Levenspiel balances and the lab-manual fitting workflow in software so students can use them *during* the three-hour slot."

3 Chemistry and the link to conductivity

Reaction (saponification):

$$\mathrm{NaOH} + \mathrm{CH_3COOC_2H_5} \rightarrow \mathrm{CH_3COONa} + \mathrm{C_2H_5OH}$$

Usually treated as $-r_A = k\,C_A C_B$. Equimolar feed $\Rightarrow -r_A = k\,C_A^2$.

Why measure conductivity? NaOH (OH^-) conducts strongly. As it is consumed, conductivity Λ falls.

$$X = \frac{\Lambda_0 - \Lambda}{\Lambda_0 - \Lambda_\infty}$$

Λ_0: feed / start. Λ_∞: complete conversion. Too-low Λ_∞ **inflates** X and **shrinks** k. App default $\approx 0.366\,\Lambda_0$; manuals often type 0.1 or 3.75 mS/cm.

4 Mathematical models (what to write on the board)

4.1 Symbols you must know

Symbol	Name	Meaning
C_A, C_B	Concentrations	NaOH, ester (mol/L)
C_{A0}	Feed concentration	After mixing, at reactor inlet
X	Conversion	Fraction of NaOH reacted
$\tau = V/Q$	Residence / space time	Tank or coil volume ÷ volumetric flow
k	Rate constant	Units depend on order (2nd: L/molůmin)
E_a, A	Activation energy, pre-factor	Arrhenius: $k = A\,e^{-E_a/RT}$
Da	Damköhler number	$k\,C_{A0}^{n-1}\tau$ — reaction speed vs flow speed
$M = C_{B0}/C_{A0}$	Stoichiometric ratio	1 = equimolar
N	Tanks in series	From RTD: $N = t_m^2/\sigma^2$
Pe	Péclet number	Convection ÷ axial dispersion

4.2 Ideal CSTR (steady, isothermal, 2nd order, equimolar)

Mole balance: $F_{A0}X = (-r_A)V$ with $-r_A = kC_{A0}^2(1-X)^2$.

$$\tau = \frac{X}{k\,C_{A0}(1-X)^2} \quad \Rightarrow \quad y = \frac{X}{(1-X)^2} = k\,C_{A0}\,\tau$$

Plot y vs τ: slope $= k\,C_{A0} \Rightarrow k$. (If $M \neq 1$, y uses $(1-X)(M-X)$ form.)

4.3 Ideal PFR (plug flow, same kinetics)

$$\tau = \int_0^X \frac{\mathrm{d}X'}{-r_A(X')} \quad \Rightarrow \quad y = \frac{X}{1-X} = k\,C_{A0}\,\tau \quad \text{(equimolar 2nd order)}$$

Along the coil the app integrates the ODE $\mathrm{d}C_A/\mathrm{d}V = r_A/Q$.

4.4 Batch

Time from concentration: $\mathrm{d}C_A/\mathrm{d}t = r_A$. Integral method: for 2nd order equimolar, $1/C_A - 1/C_{A0} = kt$. Differential method: plot $\ln(-\mathrm{d}C_A/\mathrm{d}t)$ vs $\ln C_A$; slope $\approx$ order n.

4.5 Arrhenius

$$\ln k = \ln A - \frac{E_a}{R}\,\frac{1}{T}$$

Slope of $\ln k$ vs $1/T$ is $-E_a/R$. App reports 95% intervals.

4.6 Non-ideal CSTR (speak the idea)

- **Bypass β:** fraction of feed skips the tank $\Rightarrow$ lower overall X.
- **Dead volume α:** fraction of V inactive $\Rightarrow$ smaller effective τ.
- **N tanks in series:** same total V, approaches PFR as N grows $\Rightarrow$ higher X.

4.7 Non-ideal PFR

- **Axial dispersion:** Fogler open/closed vessel; characterised by Pe. High Pe $\rightarrow$ plug flow.
- **Laminar:** velocity profile across radius; conversion below ideal plug for the same τ.
- **Sample delay:** reaction continues in the beaker; looks like a large free intercept on τ vs y.

4.8 RTD

$$E(t) = \frac{C(t)}{\int_0^\infty C\,\mathrm{d}t}, \quad t_m = \int tE\,\mathrm{d}t, \quad \sigma^2 = \int (t-t_m)^2 E\,\mathrm{d}t, \quad N = \frac{t_m^2}{\sigma^2}$$

$F(t) = \int_0^t E\,\mathrm{d}t$. Dimensionless $E(\theta)$ with $\theta = t/t_m$. Tail cut-off underestimates σ^2 and inflates N.

4.9 Energy balance (CSTR jacket)

At steady state, heat generation from reaction equals heat removal by the jacket $UA(T - T_a)$ (plus sensible heat of flow). Plot generation and removal vs T: one or three intersections $\Rightarrow$ multiple steady states / runaway.

4.10 Flowsheet (Design Studio)

Each block uses the same rate law. Mixers and splitters conserve mass. Recycle: iterate until compositions stop changing. Overall X from all feeds to product. Mass balance on Na and acetyl groups must close ($< 10^{-8}$ relative).

5 What every graph does

5.1 CSTR / PFR Results tab

Graph	What it shows / why it is there
Design plot (one T)	y vs τ (or flipped). Line through points $\rightarrow$ k at that temperature. Core of the manual.
Design plot (all T)	Same axes, several temperatures. Steeper slope = larger k.
Arrhenius	$\ln k$ vs $1/T$. Slope $\rightarrow$ E_a. Proves temperature dependence is consistent.
X vs total flow	Higher flow $\Rightarrow$ smaller τ $\Rightarrow$ usually lower X.
X vs T	Higher T $\Rightarrow$ larger k $\Rightarrow$ usually higher X.
k from each run alone	Each point is k from one row. Flat vs τ = consistent data; sloping = Λ_∞, delay, or unsteady.
Parity (model vs measured X)	Points on the diagonal = fit is good.
Residuals	Measured $-$ model for each run; shows which runs disagree.

5.2 Predictions panel

- X vs τ at chosen T: size the reactor mentally.
- k vs T: Arrhenius curve with your fitted points.
- X vs C_{A0} for orders 1–3: all orders fit at the lab C_{A0}; they split if concentration changes — that is how you tell order apart.

5.3 Assumption effects

Bar chart of X under ideal vs bypass / dead volume / N tanks / stagnant / reversible. Answers "what was that assumption costing me?"

5.4 Energy balance

Heat generation vs removal vs T; phase portrait of trajectories. Shows multiple steady states and runaway basin.

5.5 PFR: conversion along coil; flow models

$X(z/L)$: builds along the tube. Flow-models plot: same k, different hydrodynamics, points overlaid.

5.6 Batch

$C_A(t)$; integral plot (straight if order right); differential log–log (slope $\approx$ n); residuals; AIC table ranking rate laws.

5.7 RTD

$E(t)$ with t_m marked; overlay of all runs; $E(\theta)$, $F(t)$; model fits (tanks-in-series, dispersion) + residuals.

5.8 Design Studio

Profiles (C_A, T vs cumulative V); Levenspiel $1/(-r_A)$ vs X (area $\propto$ size); sweep of one parameter; $k(T)$ from every rig on one plot + ideal X–Da curves.

6 Glossary — terms the professor may ask

Residence time τ
V/Q. Average time fluid would spend if the vessel were ideally swept.

Space time
Same as τ for constant density liquid (Fogler).

Conversion X
Fraction of limiting reactant (NaOH) that has reacted.

Damköhler Da
Dimensionless rate $\times$ time. Large Da $\Rightarrow$ high X for a given reactor type.

Design plot
Linearised kinetics plot (y vs τ) whose slope gives k.

Arrhenius
How k depends on T; gives E_a.

Activation energy E_a
Barrier; higher $E_a \Rightarrow k$ rises faster with T.

Pre-exponential A
Arrhenius intercept factor (same units as k).

Equimolar
$C_{A0} = C_{B0}$ ($M = 1$).

Steady state
Properties no longer change with time. CSTR outlet conductivity flat.

Transient / start-up
Approach to steady state after start or flow change.

Bypass
Feed fraction that never enters the reacting volume.

Dead volume
Stagnant fraction of V that does not participate.

Tanks in series N
Ideal CSTRs chained; large $N \approx$ PFR.

RTD
Residence time distribution — how long tracer molecules stay.

$E(t)$
Exit age density; probability density of leaving at time t.

$F(t)$
Cumulative fraction that has left by time t.

Péclet Pe
Convective transport / axial dispersion. High Pe $\approx$ plug flow.

Levenspiel plot
$1/(-r_A)$ vs X; area related to reactor volume needed.

Parity plot
Predicted vs measured; checks the fit visually.

Residual
Measured minus model; should be small and random.

AIC
Information criterion ranking models; lower is better (batch rate laws).

Sample delay
Extra reaction time after the PFR outlet (beaker).

Multiple steady states
More than one stable T (and X) for the same feeds/jacket.

Runaway
Trajectory that jumps to the hot steady state.

UA
Jacket heat-transfer coefficient $\times$ area (cooling strength).

7 Page-by-page: what each experiment does

7.1 CSTR

Simulate / animate tank; type lab-sheet readings; extrapolate to true steady Λ; fit $k(T)$ and E_a; assumption switches; energy balance; export.

7.2 PFR

Coil profiles; average readings per flow; design plots; plug / dispersion / laminar / recycle / delay; Re and Dean from coil geometry.

7.3 Batch

$\kappa(t) \to C_A$; choose rate law; clock offset; AIC comparison; Arrhenius if several T.

7.4 RTD

Multi-run tracer sheet; moments; N and Pe (several conventions); tail warning; model ranking; conversion implied by the RTD.

7.5 Design Studio

Build series / recycle / bypass / two feeds / jacketed tank; share k from lab pages; streams, profiles, Levenspiel, sweep, compare rigs; save/load JSON.

8 3-hour lab use (students + TA)

1. Open `https://che331.streamlit.app`, name the project.
2. Set V, C_{A0}, Λ_∞ mode to match the manual.
3. Run the experiment; type readings when due (CSTR stopwatch helps).
4. Check diagnostics: settled or not? k vs τ flat?
5. Change flow/T; watch Arrhenius fill in.
6. Toggle one assumption; see ΔX.
7. Publish kinetics $\to$ Design Studio $\to$ try series/recycle.
8. Export ZIP; Project $\to$ Save JSON to share the session.

9 Numbers to remember

CSTR report match	$E_a = 25.07\,\text{kJ/mol}$; $k = 3.16/5.85/7.44$ at $31/46/60\,$řC
PFR report match	$E_a = 12.70\,\text{kJ/mol}$; $k = 11.74/13.75/15.73$ at $40/50/60\,$řC
Design Studio demo	CSTR→PFR overall $X \approx 0.545$
Default Λ_∞/Λ_0	≈ 0.366 (ions)

10 Short Q&A

Q: Did you derive new kinetics?

No — Fogler/Levenspiel mole balances and the lab-manual design plots, implemented and checked against reports.

Q: Where does k come from?

From the slope of the design plot at each T, then Arrhenius for E_a and A.

Q: What is Da?

$\text{Da} = kC_{A0}^{n-1}\tau$. Large Da means reaction is fast compared with the time fluid spends in the reactor.

Q: Why can CSTR and PFR E_a differ between groups?

Different Λ_∞, scatter, sample delay, unsteady averages. App matches each report's method; fair comparison needs the same conventions (Compare rigs).

Q: Best insight from the tool?

Low typed Λ_∞ and averaging before steady state explain weird numbers better than "wrong kinetics."

Know the pitch, the design-plot equation, Arrhenius, and what each Results graph is for. That is enough to explain the whole project.